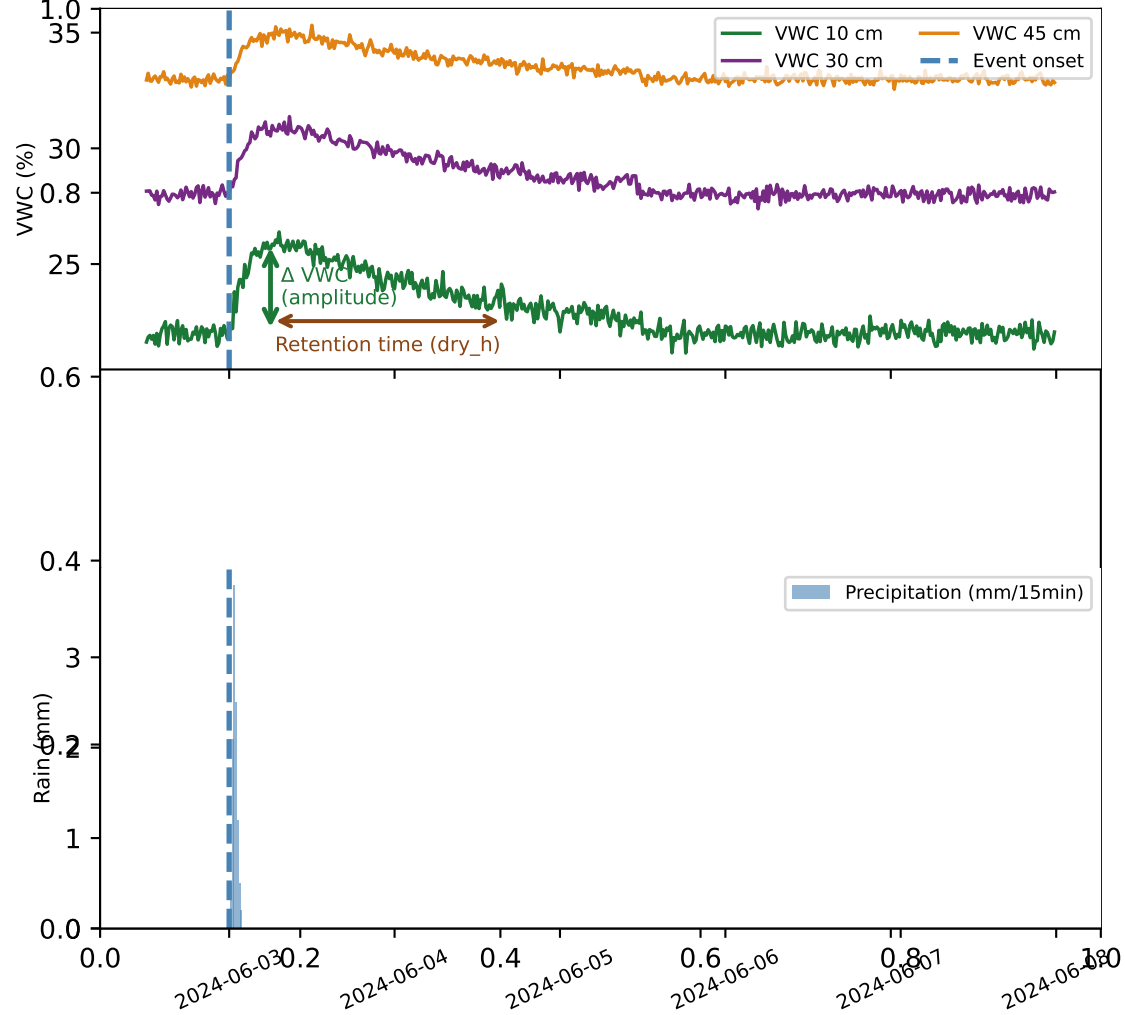
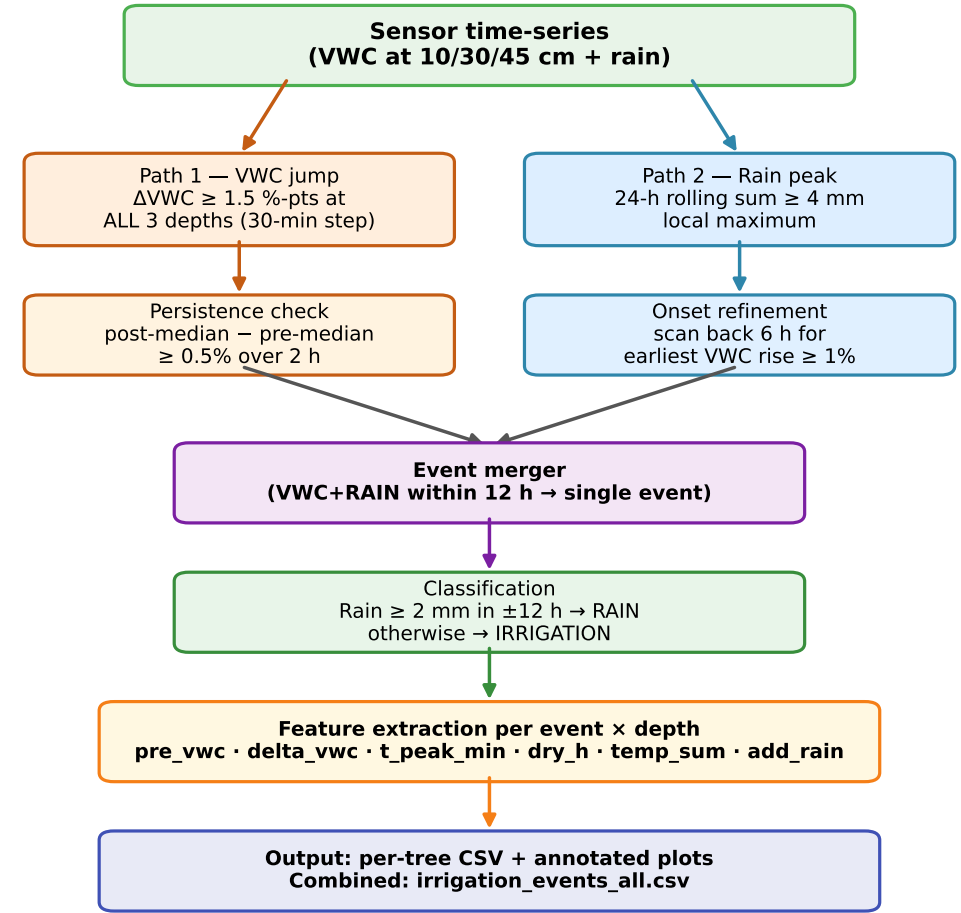


Precipitation and Irrigation Event Detection Framework

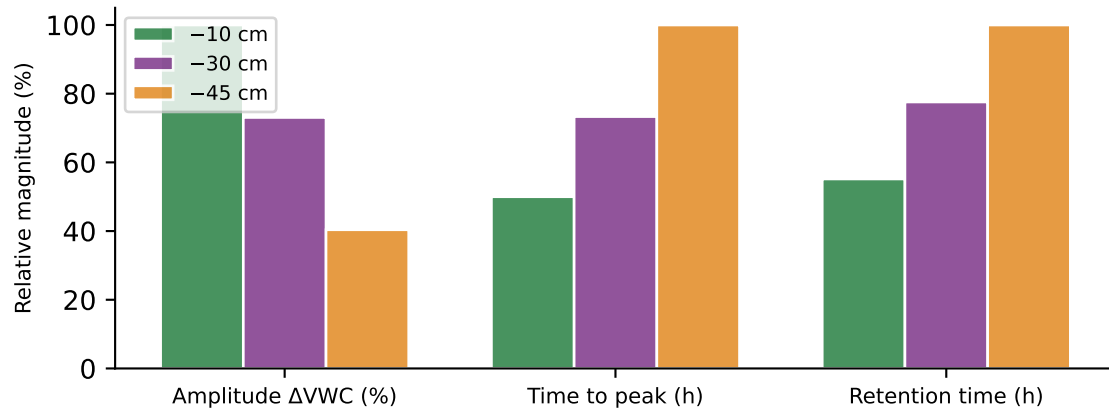
(a) Example precipitation event: soil moisture response



(b) Two-path event detection algorithm



(c) Extracted event features by depth (schematic)



Key parameters:
JUMP_THR = 1.5 %/30 min
PERSIST_H = 2 h
RAIN_EVENT_MIN = 4 mm
EVENT_MERGE_H = 12 h
DRY_MAX_H = 168 h (7 d)
DRYING_FRAC = 10%